Movie Ticket Reservation System: Requirements Engineering Document

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Abstract. This document presents the requirements engineering artifacts for a Movie Ticket Reservation System developed in C++ as part of the FSOFT course at ISEP. The system allows users to register an account, log in, check movie schedules and prices, and make reservations by choosing a movie, specifying the number of tickets, selecting chairs, and entering payment details. The document covers Requirements Engineering concepts, User Stories, Domain Model, Functional and Non-Functional Requirements (FURPS+), Use-Case Model with specifications, System Sequence Diagrams, and a Glossary.

Keywords: Requirements Engineering, User Stories, Use Cases, Domain Model, FURPS+, SSD, Glossary, Movie Ticket Reservation, C++

1 Introduction

This document defines and documents the requirements for a Movie Ticket Reservation System to be implemented in C++. The system provides the following main functionalities to its users:

Register and Login: Users create an account and authenticate before accessing system features.

Check Schedule: Users can view the schedule of available movies, including dates, times, and screening rooms.

Check Prices: Users can consult ticket prices for different movies and seat types.

Reservation: Users follow a guided flow: choose a movie, specify the number of tickets, select chairs from the seating map, enter payment details, and receive a reservation confirmation screen.

Exit: Users can log out and exit the system.

The requirements engineering process follows the methodology taught in FSOFT, producing the following artifacts: Requirements Engineering overview, User Stories, Domain Model, Functional and Non-Functional Requirements (FURPS+), Use-Case Model, System Sequence Diagrams, and a Glossary.

2 User Stories

User stories are business needs, not requirements in the traditional sense. They are oriented toward the user and a business need. A user story describes a business need, not the system's functionality — the functionality that fulfills the need is the Development Team's job. A user story is an informal, general explanation of a software feature written from the perspective of the end user, following the format: As a <role>, I want to <action>, so that <benefit>.

Each story includes acceptance criteria — detailed conditions that a feature must meet to be considered complete. Acceptance criteria are more technical, offering a checklist that ensures the feature behaves as intended. They should be testable and reveal straightforward yes/no or pass/fail results.

2.1 US01 – Register

User Story	As a new user, I want to register an account so that I can access the reservation system.

Acceptance Criteria	1. The system requests a username, email, and password. 2. The password must have a minimum of 8 characters, at least 1 uppercase, 1 lowercase, and 1 digit. 3. The system validates that the username and email are not already in use. 4. Upon successful registration, the system confirms the account creation.

2.2 US02 – Login

User Story	As a registered user, I want to log in so that I can access the system features.
Acceptance Criteria	1. The system requests username and password. 2. The system validates the credentials against the stored data. 3. Upon successful login, the system displays the main menu. 4. After 3 failed attempts, the account is temporarily locked.

2.3 US03 – Check Schedule

User Story	As a user, I want to check the movie schedule so that I can see what movies are available and when.
Acceptance Criteria	1. The system displays a list of movies with their showtimes.
	2. Each entry shows the movie title, date, time, and screening room. 3. Only future showtimes are displayed.

2.4 US04 – Check Prices

User Story	As a user, I want to check ticket prices so that I can decide which movie and seat type fits my budget.
Acceptance Criteria	1. The system displays the price for each movie. 2. Prices are shown per seat type (e.g., regular, VIP). 3. Any active promotions or discounts are displayed.

2.5 US05 – Make a Reservation

User Story	As a user, I want to make a reservation so that I can guarantee my seats for a movie.

Acceptance Criteria	1. The user selects a movie from the available list. 2. The user specifies the number of tickets. 3. The system displays the seating map for the chosen screening. 4. The user selects the desired chairs from available seats. 5. The system requests payment details (card number, expiry, CVV). 6. Upon successful payment, a reservation confirmation screen is displayed with a unique booking reference. 7. The reserved chairs are marked as unavailable for other users.

2.6 US06 – Exit

User Story	As a user, I want to exit the system so that I can safely end my session.
Acceptance Criteria	1. The system logs the user out. 2. Any unsaved reservation in progress is discarded. 3. The system returns to the login/welcome screen.

3 Requirements Engineering

3.1 What is a Requirement?

A requirement is a necessary attribute in a system, a statement that identifies a capability, characteristic, or quality factor of a system in order for it to have value and utility to a customer or user. Requirements are important because they provide the basis for all of the development work that follows. Once the requirements are set, developers initiate the other technical work: system design, development, testing, implementation, and operation.

A requirement is a condition or capability required by a stakeholder to solve a problem or achieve an objective, while a need is a high-level representation of the requirement. The need is the end result or purpose — it is the why we are doing this.

3.2 Requirement vs Need

The following table illustrates the distinction between a need and the requirements that fulfill it, applied to our Movie Ticket Reservation System:

Need (I need to...)	Requirements
Reserve seats for a movie online	The system shall display a list of available movies.
	The system shall show available showtimes for a selected movie.
	The system shall display a seating map with available and occupied chairs.
	The system shall allow the user to select specific chairs.
	The system shall request payment details to confirm the reservation.
	The system shall generate a unique booking reference.
	The system shall display a reservation confirmation screen.

4 Domain Model

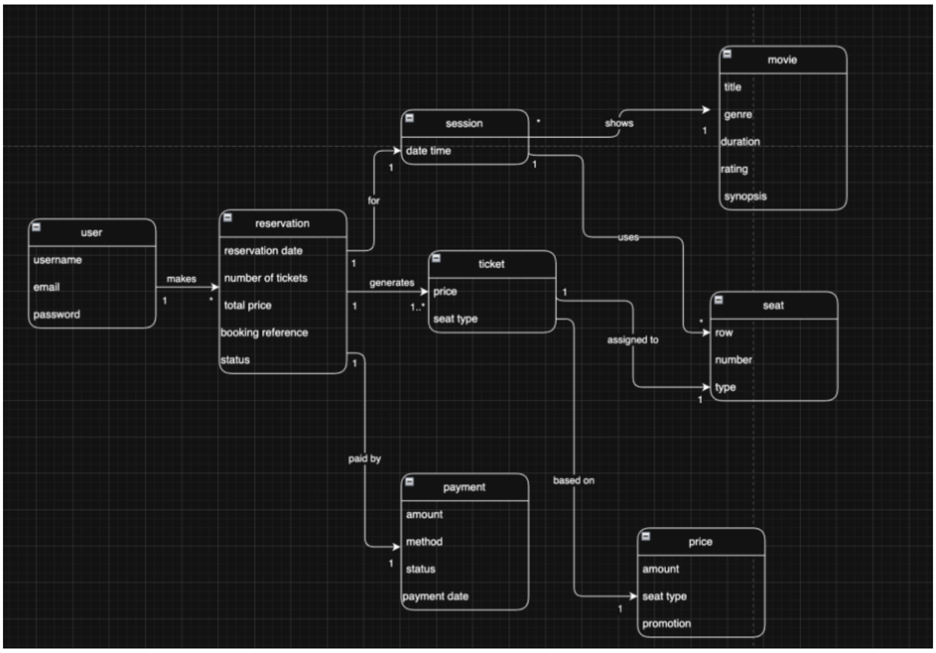

Figure 1- Domain Model

5 Requirements Specification

Requirements are classified into functional and non-functional categories. Functional requirements describe what the system must do. Non-functional requirements are categorized using the FURPS+ model.

5.1 Functional Requirements

ID	Description
FR01	The system shall allow a new user to register with username, email, and password.
FR02	The system shall validate that username and email are unique during registration.
FR03	The system shall enforce password rules (min. 8 chars, 1 uppercase, 1 lowercase, 1 digit).
FR04	The system shall allow a registered user to log in with username and password.
FR05	The system shall lock the account after 3 consecutive failed login attempts.
FR06	The system shall display the movie schedule with title, date, time, and room.
FR07	The system shall allow filtering the schedule by date or movie title.
FR08	The system shall display ticket prices per movie and seat type.
FR09	The system shall allow the user to select a movie for reservation.
FR10	The system shall allow the user to specify the number of tickets.
FR11	The system shall display the seating map showing available and occupied chairs.
FR12	The system shall allow the user to select specific chairs from available seats.
FR13	The system shall request payment details (card number, expiry date, CVV).
FR14	The system shall validate payment details before processing.
FR15	The system shall generate a unique booking reference upon successful reservation.
FR16	The system shall display a reservation confirmation screen with booking details.
FR17	The system shall mark reserved chairs as unavailable for other users.
FR18	The system shall allow the user to exit and log out safely.

5.2 Non-Functional Requirements (FURPS+)

ID	Category	Description

NFR01	Functionality	The system shall require login before accessing schedule, prices, or reservation features.
NFR02	Functionality	The system shall store user passwords in an encrypted format.
NFR03	Functionality	The system shall log all reservation transactions for auditing.
NFR04	Usability	The seating map shall visually distinguish available, occupied, and selected chairs.
NFR05	Usability	The system shall provide clear error messages for invalid inputs.
NFR06	Usability	The main menu shall clearly present all options: Check Schedule, Check Prices, Reservation, Exit.
NFR07	Reliability	The system shall not lose confirmed reservation data in case of unexpected termination.
NFR08	Reliability	The system shall handle concurrent seat selections to prevent double-booking.
NFR09	Performance	The system shall display the schedule and prices without delay.
NFR10	Performance	The seating map shall update in real time after a chair is selected.
NFR11	Supportability	The system shall be developed in C++ with modular architecture.
NFR12	Supportability	The system shall include unit tests for registration, login, and reservation logic.

Additional Constraints (+). The + in FURPS+ covers design, implementation, interface, and physical constraints.

Constraint Type	Description
Design	The system shall follow a layered architecture (UI, business logic, data persistence).
Implementation	The system shall be implemented in C++ using standard libraries.
Interface	The system shall use a console-based (text) user interface.
Physical	The system shall run on standard desktop hardware (Windows/Linux).

6 Use-Case Model

Use case diagrams are used to gather usage requirements of a system and to identify functions and how roles interact with them. The use-case model identifies the actors interacting with the system and the use cases representing system functions.

6.1 Actors

An actor in a use case diagram is any entity that performs a role in one given system. This could be a person, organization, or an external system.

Actor	Description
Unregistered User	A person who has not yet created an account. Can only register.
User	A registered and logged-in person who can check schedules, prices, make reservations, and exit.

6.2 Use Cases

A use case represents a function or an action within the system. The system (or subject) is used to define the scope of the use case. An association is the relationship between an actor and a use case, indicating that an actor can perform a certain action.

ID	Use Case	Actor(s)
UC01	Register	Unregistered User
UC02	Login	Unregistered User
UC03	Check Schedule	User
UC04	Check Prices	User
UC05	Make a Reservation	User
UC06	Exit	User

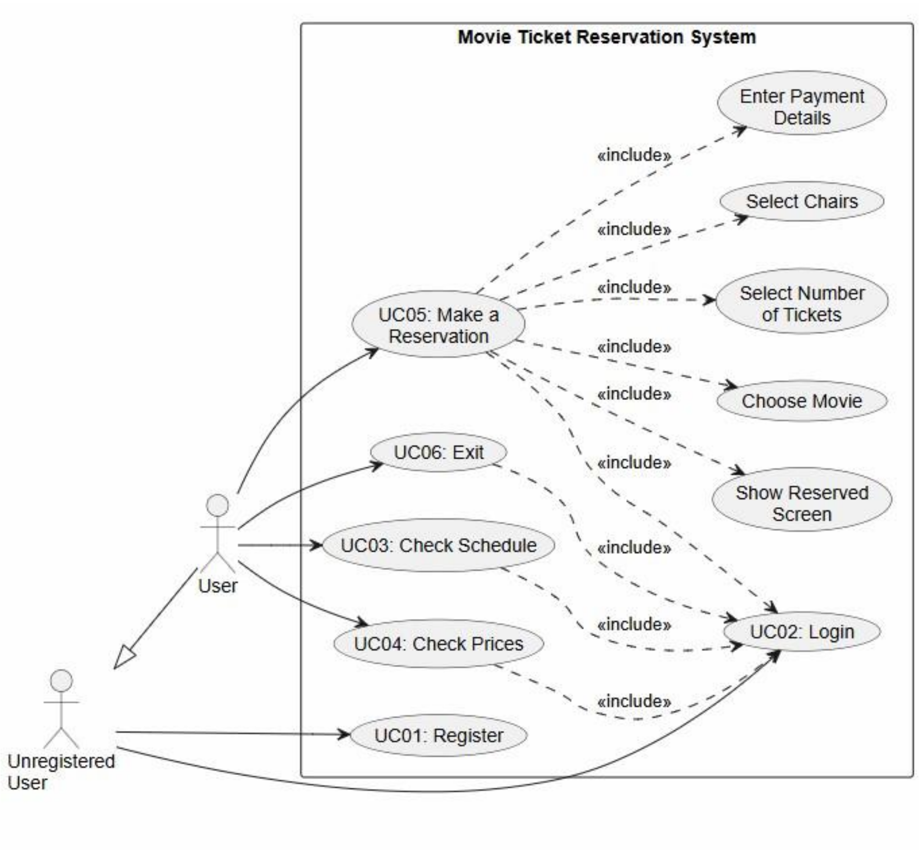

Figure 2 - Use Case Model

6.3 Use Case Specification – UC01: Register

Actor	Unregistered User
Use Case Name	Register
Description	Allows a new user to create an account to access the system.
Precondition	The user does not have an existing account.
Postcondition	A new user account is created and stored in the system.
Main Flow	1. The system displays the registration form. 2. The user enters a username, email, and password. 3. The system validates that the username and email are unique. 4. The system validates the password meets the rules (min. 8 chars, 1 upper, 1 lower, 1 digit). 5. The system creates the account. 6. The system displays a success message and redirects to the login screen.
Alternative Path	3a. The username or email is already in use. → The system displays an error. Resume at step 2. 4a. The password does not meet the requirements. → The system displays an error. Resume at step 2.
Exceptions	At any point, the user may cancel the registration. → The system returns to the welcome screen.

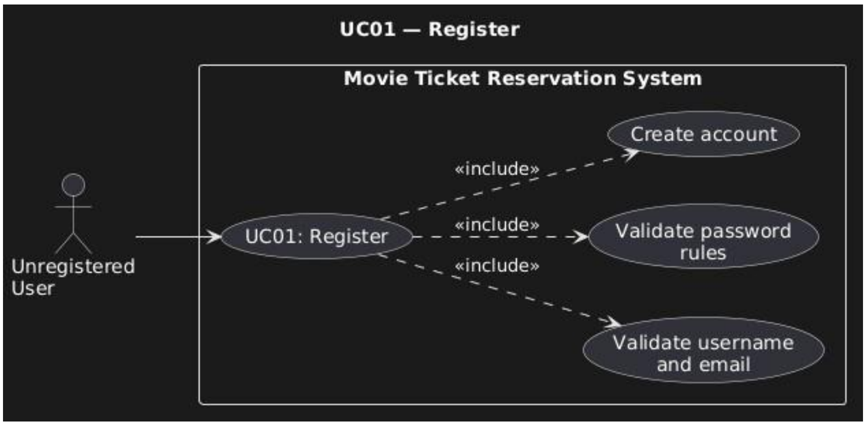

Figure 3 – UC01 Register

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6.4 Use Case Specification UC02: Login

Actor	Unregistered User (with existing account)
Use Case Name	Login
Description	Allows a registered user to authenticate and access the system.
Precondition	The user has a registered account.
Postcondition	The user is authenticated and the main menu is displayed.
Main Flow	1. The system displays the login form. 2. The user enters username and password. 3. The system validates the credentials. 4. Credentials are valid. 5. The system displays the main menu (Check Schedule, Check Prices, Reservation, Exit).
Alternative Path	3a. The credentials are invalid. → The system displays an error. The failed attempt counter increments. Resume at step 1.
Exceptions	At any point, the user may cancel. → The system returns to the welcome screen.

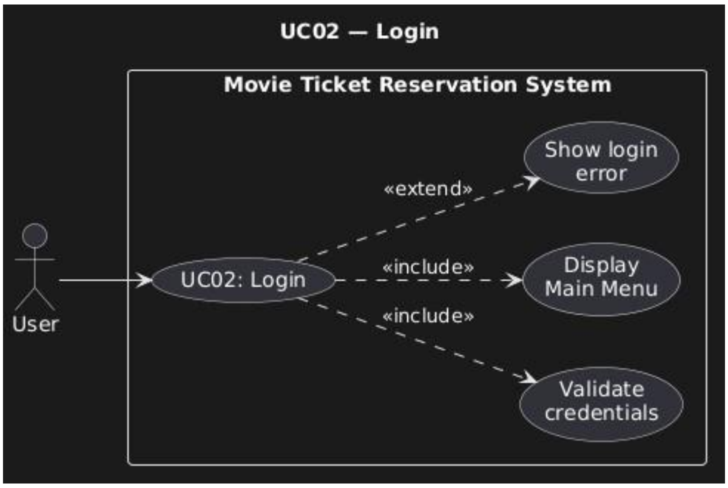

Figure 4 – UC02 Login

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6.5 Use Case Specification UC03: Check Schedule

Actor	User (logged in)
Use Case Name	Check Schedule
Description	Allows the user to view the schedule of available movies with dates, times, and screening rooms.
Precondition	The user is logged in.
Postcondition	The movie schedule is displayed to the user.
Main Flow	1. The user selects Check Schedule from the main menu. 2. The system retrieves all future movie screenings. 3. The system displays a list of movies with date, time, and screening room. 4. The user views the schedule. 5. The user returns to the main menu.
Alternative Path	3a. No screenings are available. → The system displays a message indicating no movies are currently scheduled.

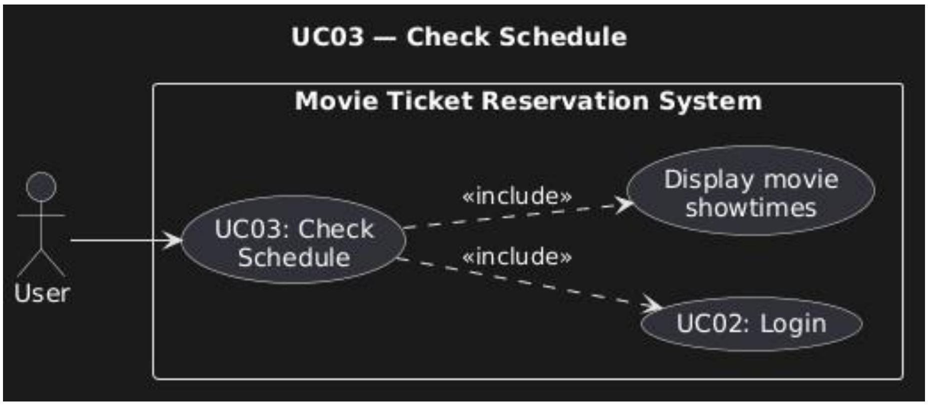

Figure 5 - UC03 Check Schedule

6.6 Use Case Specification – UC04: Check Prices

Actor	User (logged in)
Use Case Name	Check Prices
Description	Allows the user to consult ticket prices for different movies and seat types.
Precondition	The user is logged in.
Postcondition	Ticket prices are displayed to the user.
Main Flow	1. The user selects Check Prices from the main menu. 2. The system retrieves the price list for all movies. 3. The system displays prices per movie and per seat type (regular, VIP). 4. The system displays any active promotions or discounts. 5. The user views the prices. 6. The user returns to the main menu.
Alternative Path	3a. No promotions are currently active. → The system displays prices without a promotions section.
Exceptions	At any point, the user may return to the main menu.

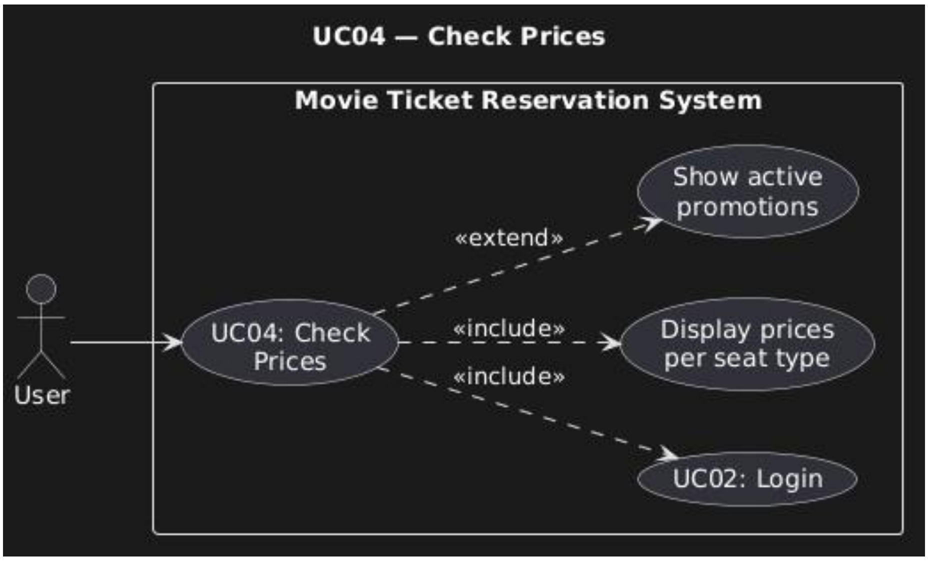

Figure 6 - UC04 Check Prices

6.7 Use Case Specification – UC05: Make a Reservation

Actor	User (logged in)
Use Case Name	Make a Reservation
Description	Allows the user to reserve seats for a movie through a guided flow: choose movie, specify tickets, select chairs, enter payment, and receive confirmation.
Precondition	The user is logged in. There are movies with available screenings and seats.
Postcondition	A reservation is created with a unique booking reference. The selected chairs are marked unavailable.

Main Flow	1. The system displays the list of available movies. 2. The user selects a movie (Choose Movie). 3. The system displays available screenings for the selected movie. 4. The user selects a screening (date/time). 5. The system asks how many tickets the user wants (How Many Tickets). 6. The user enters the number of tickets. 7. The system displays the seating map with available and occupied chairs (Select Chairs). 8. The user selects the desired chairs (must match the number of tickets). 9. The system displays the order summary (movie, screening, chairs, total price). 10. The system requests payment details: card number, expiry date, CVV (Payment Details). 11. The user enters the payment information. 12. The system validates the payment details. 13. The system processes the payment. 14. The system creates the reservation with a unique booking reference. 15. The system marks the selected chairs as unavailable. 16. The system displays the reservation confirmation screen (Reserved Screen).
Alternative Path	8a. The user selects more or fewer chairs than the number of tickets. → The system displays an error. Resume at step 8. 8b. One or more selected chairs are no longer available. → The system updates the seating map. Resume at step 7. 12a. The payment details are invalid (e.g., wrong format). → The system displays an error. Resume at step 10. 13a. The payment is declined. → The system informs the user. Resume at step 10.
Exceptions	At any point, the user may cancel the reservation. → Any temporarily held chairs are released. → The system returns to the main menu.

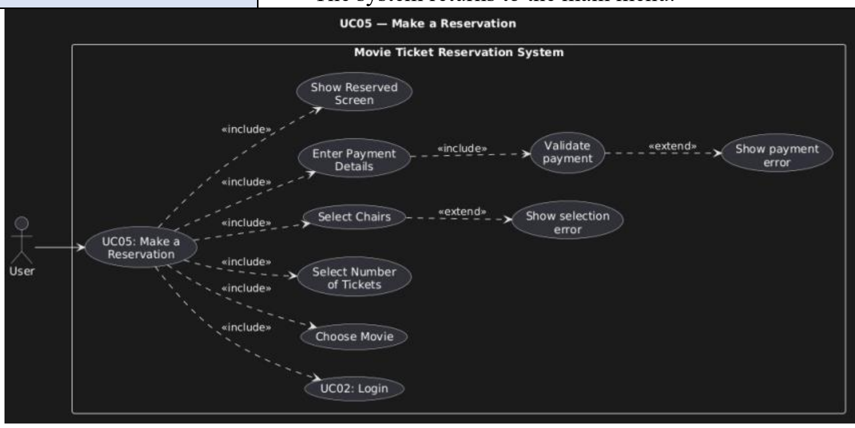

Figure 7 - UC05 Make a Reservation

6.8 Use Case Specification – UC06: Exit

Actor	User (logged in)
Use Case Name	Exit
Description	Allows the user to safely log out and terminate the session.
Precondition	The user is logged in.
Postcondition	The user is logged out and the system returns to the welcome screen.
Main Flow	1. The user selects Exit from the main menu. 2. The system logs the user out. 3. The system discards any unsaved reservation in progress. 4. The system returns to the welcome/login screen.
Alternative Path	2a. A reservation is currently in progress. → The system asks the user to confirm they want to exit and discard the reservation. → If confirmed, the reservation is discarded and chairs are released. Resume at step 2. → If not confirmed, the system returns to the main menu.
Exceptions	

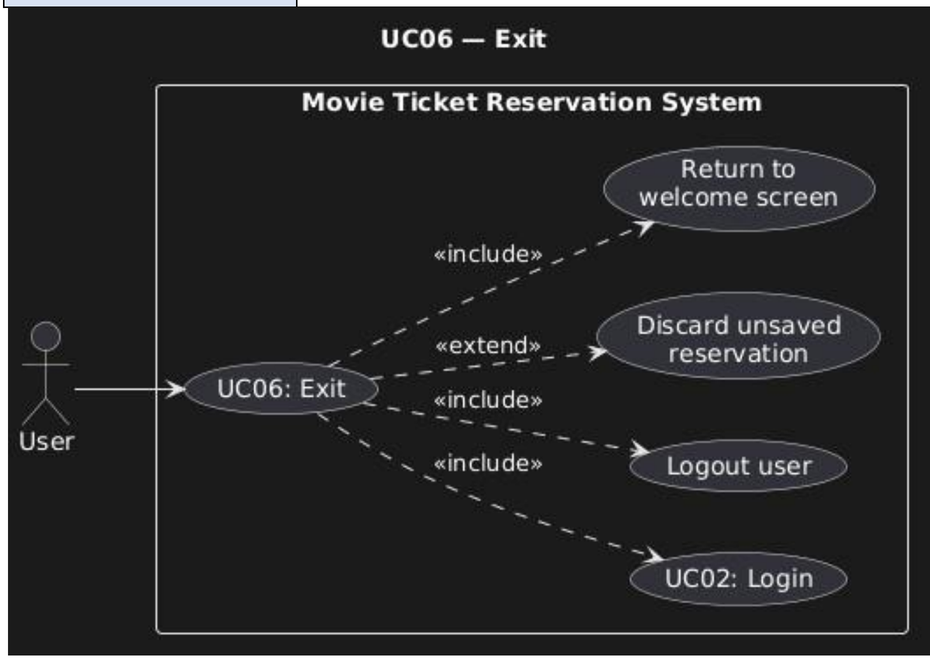

Figure 8 - UC06 Exit

7 Glossary

Term (EN)	Termo (PT)	Description
Account	Conta	A user record in the system containing username, email, and password.
Booking Reference	Referência de Reserva	A unique alphanumeric code identifying a reservation.
Chair	Cadeira/Lugar	An individual seating position in a screening room, identified by row and number.
CVV	CVV	Card Verification Value. A 3-digit security code on the payment card.
Exit	Sair	The action of logging out and terminating the user session.
Genre	Género	A category classifying a movie (e.g., Action, Comedy, Drama).
Login	Iniciar Sessão	The process of authenticating a user with username and password.
Main Menu	Menu Principal	The screen displayed after login with options: Schedule, Prices, Reservation, Exit.
Movie	Filme	A film available for screening at the theater.
Payment	Pagamento	A financial transaction using card details to confirm a reservation.
Price	Preço	The monetary cost of a ticket, which may vary by seat type.
Rating	Classificação	An age-based classification assigned to a movie (e.g., PG, PG-13, R).
Registration	Registo	The process of creating a new user account in the system.
Reservation	Reserva	A confirmed booking of specific chairs for a movie screening.
Reserved Screen	Ecrã de Confirmação	The confirmation screen shown after a successful reservation.
Schedule	Horário	The list of movie screenings with dates, times, and rooms.
Screening Room	Sala de Cinema	A physical room in the theater where movies are projected.
Seating Map	Mapa de Lugares	A visual layout of chairs in a screening room showing availability.
SSD	SSD	System Sequence Diagram. A UML diagram showing actor-system interaction.
Ticket	Bilhete	A purchased item granting access to a specific chair at a screening.
UC	UC	Use Case. A description of a system function triggered by an actor.
User	Utilizador	A registered person who can access system features after login.
VIP	VIP	A premium seat type with additional comfort, typically at a higher price.

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